

# Nikhil Kumar

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## SUMMARY

Mechanical Engineer (Ph.D.) specializing in wheel-rail contact mechanics, multi-body dynamics simulation, and experimental vehicle dynamics research, with a sustained record of federally sponsored work for the FRA. Demonstrated ability to lead original experimental programs, mentor junior researchers, and translate complex technical findings into peer-reviewed publications and policy-relevant guidance for federal stakeholders. Seeking a faculty position to build an independent research program in rail vehicle dynamics and safety.

## RESEARCH & WORK EXPERIENCE

### Postdoctoral Associate | *Non-tenure Track Research Faculty* | CVeSS Lab (VT)

2025 - Present

PI: Dr. Mehdi Ahmadian

Blacksburg, VA

- Designed and built **electromechanical test infrastructure**, expanding roller rig capability from single-wheel to single-axle testing.
  - Specified servo motors, drives, and integrated 3 new axes into existing control architecture — bridging hardware to control model.
  - Developed and implemented lab safety protocols (STO, E-Stop, RTO) ensuring compliance with operational safety standards.
  - Stress-tested and calibrated custom PCB for wheel instrumentation, validating signal integrity under operational loads.
  - Adding multiple new single and gantry axes into an existing control architecture to provide a bridge between the drives and the control model.

### PhD Candidate | *Graduate Research Assistant* | CVeSS Lab (VT) Dr. Mehdi Ahmadian (Advisor)

2022 - 2025

Blacksburg, VA

#### Dissertation title: Effect of Wheel and Rail Conditions on Contact Dynamics and Mechanics

- Designed and executed structured experimental protocols evaluating solid and liquid contaminant effects on wheel-rail tractive forces — findings delivered to the Federal Railroad Administration.
- Characterized complex non-Hertzian contact conditions (conformal and two-point) using AAR2A wheel profile — first experimental study of its kind in the field
- Applied JMP-based DOE across multiple degrees of freedom; produced statistically rigorous technical reports for federal stakeholders

### Undergraduate Research Assistant | Dr. K Anandavel (Advisor)

2020 - 2022

Vellore, India

- FEA contact modeling of stepped and back-to-back bolted joints using ANSYS APDL; parametric evaluation of stress/strain at frictional interfaces
- Received "Best Paper" award at peer-reviewed conference

## RELEVANT ACADEMIC PROJECTS

### VT-FRA Rig Wear Optimization at Wheel-Rail Contact Interface, Spring 2023

- Developed an optimization problem to minimize wear rate using constrained unconstrained equations.

- Achieved successful minimization using Sequential Quadratic Programming (SQP) for efficiently handling nonlinear constraints used on the modified Archard's wear law.
- Results validated with experimental baseline testing done on the VT-FRA Roller Rig shows a 95% reduction in rate of wear generated.

### **Design for Additive Manufacturing (DfAM) of a Motorcycle Seat, Fall 2023**

- Additive-manufactured motorcycle seat for customized comfort.
- Designed and implemented a seat based on topological optimization using Polypropylene material on the ProMaker P2000HT printer.
- Performed various cost analysis and feasibility testing using DfAM approaches.

### **A Study on Frictional Effect at Contact Interfaces of Aero-Engine Bolted Joints, Fall 2020**

- Design and FEM analysis of stepped and back-to-back flanges using ANSYS Mechanical Workbench and APDL.
- Designed parametric evaluation code using APDL for faster prediction time.

## **SKILLS**

| Management                       | Simulations and FEA               | Data and Instrumentation     | Research Interests      |
|----------------------------------|-----------------------------------|------------------------------|-------------------------|
| • Leadership                     | • SolidWorks, Fusion 360, AutoCAD | • MATLAB                     | • Mechanical Design     |
| • Communication and Adaptability | • Ansys                           | • Python                     | • Vehicle Dynamics      |
| • Time Management                | • Abaqus                          | • JMP                        | • Experimental Analysis |
| • Technical Report Writing       | • CONTACT                         | • Minitab                    | • Contact Mechanics     |
| • Stakeholder Presentations      | • Mechanical Tools                | • Experimental Test Protocol | • Signal Processing     |

## **PUBLICATIONS & SCHOLARLY CONTRIBUTIONS**

### **Peer-reviewed Journals**

1. **Nikhil Kumar**, Ahmad Radmehr, and Mehdi Ahmadian. 2024. "**Experimental Evaluation of Effect of Leaves on Railroad Tracks in Loss of Braking**" *Machines* 12, no. 5: 301. <https://doi.org/10.3390/machines12050301>
2. **Nikhil Kumar**, Ahmed Radmehr, and Mehdi Ahmadian. 2025. "**Traction enhancement evaluation of Magnetite and Alumina**" *International Journal of Rail Transportation*, 1–19. <https://doi.org/10.1080/23248378.2025.2484807>
3. Giovanni Mantovani, **Nikhil Kumar**, and Mehdi Ahmadian. 2025. "**Design Evaluation of a Single Wheelset Roller Rig for Railroad Curving Dynamics and Creepage Studies**" *Designs* 9, no. 4: 99. <https://doi.org/10.3390/designs9040099>
4. Aashish Shaju, **Nikhil Kumar**, Giovanni Mantovani, Steve Southward, and Mehdi Ahmadian, 2025 "**Practical Application of Passive Air-Coupled Ultrasonic Acoustic Sensors for Wheel Crack Detection,**" *Sensors* 25, no. 19: 6126. <https://doi.org/10.3390/s25196126>
5. **Nikhil Kumar**, Mehdi Ahmadian, and Brian Marquis. "**Wheel-Rail Induced Derailment Analysis: A Comprehensive Literature Review of Experimental and Simulation-Based Approaches,**" *Railway Engineering Science*. <https://link.springer.com/article/10.1007/s40534-025-00425-0>

6. **Nikhil Kumar**, Giovanni Mantovani, and Mehdi Ahmadian. **“Dynamic Experimental Characterization of Conformal and Two-Point Wheel–Rail Contact on the VT–FRA Roller Rig”** [Submitted]

### Conference Proceedings

1. **Nikhil Kumar**, Giovanni Mantovani, Ahmad Radmehr, and Mehdi Ahmadian. 2024, **“Estimation of Wheel–Rail Contact Patch Geometry Under Curving Conditions”** Proceedings of the 2024 American Railway Engineering and Maintenance-of-Way Association Annual Conference.  
<https://doi.org/10.5281/zenodo.16055410>
2. Giovanni Mantovani, **Nikhil Kumar**, Ahmad Radmehr, and Mehdi Ahmadian. 2025, **“Development of a Wheelset Roller Rig for Improving Rolling Stock Traction Efficiency”** Proceedings of the 2025 American Railway Engineering and Maintenance-of-Way Association Annual Conference.  
<https://doi.org/10.5281/zenodo.17219621>
3. **Nikhil Kumar**, Bhuvesh Kaushik, Anandavel Kaliyaperumal. 2020, **“A Study on Frictional Effect at Contact Interfaces of Aero–Engine Bolted Joints”** Proceedings of the 2020 Janana Chillume National Conference of Jain University. <https://doi.org/10.5281/zenodo.5787933>

### Poster and Technical Presentation

1. **Poster** “Effect of Traction Reducing and Enhancing Agents on Wheel–Rail Contact Interface” – Walter O'Brien Graduate Research Symposium – 2023
2. **Poster** “Traction reducing effect of leaves at Wheel–Rail Contact Interface” – Walter O'Brien Graduate Research Symposium – 2024
3. **Invited Talk** “Traction reducing effect of leaves at Wheel–Rail Contact Interface” – PSR Congress, UNLV, Las Vegas – 2024
4. **Poster** “Experimental Evaluation of Effect of Leaves on Railroad Tracks in Loss of Braking” – Big Data Conference, University of Delaware, 2024
5. **Pitch & Poster** “Estimation of Wheel–Rail Contact Patch Geometry Under Curving Conditions” – Walter O'Brien Graduate Research Symposium – 2025
6. **Workshop Presentation** “Influences of Contact Geometry on WRI Pressure Distribution” – FRA 2026 Workshop on Advancing VTI Simulation Capabilities for Enhanced Railroad Safety and Performance

### Scholarly Review Contributions

- Reviewer for *SAE Technical Papers*
- Reviewer for *International Journal of Rail Transportation (IJRT)*
- Reviewer for the *Railway Engineering Sciences (Springer Nature)*

## EDUCATION

| Virginia Polytechnic Institute and State University                                                                                                                                                                       | Vellore Institute of Technology                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Ph.D. Mechanical Engineering, 2022 – Present</li> <li>• GPA: 3.91/4.0</li> <li>• <b>Thesis:</b> Effect of Wheel and Rail Conditions on Contact Dynamics and Mechanics</li> </ul> | <ul style="list-style-type: none"> <li>• B.Tech. Mechanical Engineering, 2018 – 2022</li> <li>• GPA: 8.56/10.0</li> <li>• <b>Senior year project:</b> A systematic process improvement in tooling at 3C machining lines using the Toyota Production System – Toyota Engines India Pvt Ltd.</li> </ul> |

VOLUNTEERING SERVICE AND OUTREACH

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|--------------------------------------------------------------------------------|-------------|
| <b>President</b>                                                               | 2025 - 2026 |
| <i>VT – AREMA Student Chapter, Blacksburg, VA</i>                              |             |
| <b>Vice President</b>                                                          | 2024 - 2025 |
| <i>Mechanical Engineering Graduate Student Council (MEGSC), Blacksburg, VA</i> |             |
| <b>Student Mentor</b>                                                          | 2024 - 2025 |
| <i>Department of Mechanical Engineering, Virginia Tech, Blacksburg, VA</i>     |             |
| <b>Finance Head</b>                                                            | 2020 – 2021 |
| <i>Becoming I Foundation, Vellore, INDIA</i>                                   |             |

RELEVANT COURSEWORKS

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Applied Numerical Methods | Vehicle Dynamics | Finite Element in Machine Design | Machining Processes and Metrology | Solid Mechanics | Operations Research | Applied Linear Systems | Materials Engineering and Technology | Total Quality Management and Reliability | Lean Enterprises and New Manufacturing Technology | Additive Manufacturing | Digital Signal Processing | Mechanics of Materials | Micro Electromechanical Systems | Statistics in Research